Binary and Computer Systems

Understand how digital systems represent information with bits.

Beginner lesson · Estimated reading time: 8–12 minutes

Introduction

Computers use physical states to represent information. At the logical level, binary digits (bits) take values 0 and 1. Groups of bits encode numbers, text, images, and instructions according to agreed rules. Learning the representation helps explain memory, storage, and digital logic.

Learning objectives

• Explain bits, bytes, and positional notation.

• Convert small positive integers between decimal and binary.

• Perform basic binary addition.

• Distinguish data representation from the meaning assigned to it.

1. Bits, bytes, and digital states

A bit is a binary digit: 0 or 1. A byte is commonly 8 bits, allowing 2■ = 256 possible bit patterns. A pattern does not carry meaning by itself; a format determines whether it represents a number, a character, a color component, or something else.

Digital circuits use physical signals with defined ranges interpreted as logical 0 or 1. Real systems must tolerate noise and timing limits; binary representation is an abstraction that makes reliable processing practical.

2. Positional notation

In base 10, each digit's place represents a power of 10. In base 2, each place represents a power of 2. From right to left, binary place values are 1, 2, 4, 8, 16, and so on.

Binary 1011■ = 1×8 + 0×4 + 1×2 + 1×1 = 11■■. The subscript indicates the base and avoids ambiguity.

3. Convert decimal to binary

Repeatedly divide the positive decimal number by 2 and record each remainder. Read the remainders from bottom to top. Example: 13 ÷ 2 gives remainder 1; 6 ÷ 2 gives 0; 3 ÷ 2 gives 1; 1 ÷ 2 gives 1. Reading upward gives 1101■.

You can check the result by expanding it: 1101■ = 8 + 4 + 0 + 1 = 13■■. For zero, the representation is 0■.

4. Binary addition

The basic rules are 0+0=0, 0+1=1, 1+0=1, and 1+1=10■ (write 0 and carry 1). With a carry, 1+1+1=11■ (write 1 and carry 1).

Example: 101■ + 011■ = 1000■. In decimal, 5 + 3 = 8, so the result is consistent. Always account for carries from the previous column.

5. Where representations are used

Unsigned n-bit binary can represent integers from 0 through 2■–1. Four bits represent 16 patterns, from 0000 to 1111. Signed integers require a convention, such as two's complement; floating-point numbers use a different encoding. Text uses character encodings, and images use pixel/color formats.

Do not assume every bit pattern is an ordinary positive integer. The same eight bits can have different meanings under different data types and encodings.

6. Common mistakes